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Analisi dinamica di base

Consegna: “Rispondere ai seguenti quesiti, con riferimento al file eseguibile notepad-classico.exe:

15:20:54,3805440	calcolatriceinnovativa.exe	4480	Process Start		SUCCESS	Parent PID: 3532, Command line: "C:\Users\user\Desktop\Malware\ca
15:20:54,3805463	calcolatriceinnovativa.exe	4480	Thread Create		SUCCESS	Thread ID: 3692
15:20:54,4152595	calcolatriceinnovativa.exe	4480	Load Image	C:\Users\user\Desktop\Malware\calcolatriceinnovativa.exe	SUCCESS	Image Base: 0x1000000, Image Size: 0x1f000
15:20:54,4153345	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\System32\ntdll.dll	SUCCESS	Image Base: 0x7ff9f2110000, Image Size: 0x1c2000
15:20:54,4153996	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\ntdll.dll	SUCCESS	Image Base: 0x77120000, Image Size: 0x179000
15:20:54,4166035	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\System32\wow64.dll	SUCCESS	Image Base: 0x72320000, Image Size: 0x4f000
15:20:54,4169341	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\System32\wow64win.dll	SUCCESS	Image Base: 0x72370000, Image Size: 0x73000
15:20:54,4182686	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\System32\kernel32.dll	SUCCESS	Image Base: 0x140000, Image Size: 0xad000
15:20:54,4184000	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\kernel32.dll	SUCCESS	Image Base: 0x76f70000, Image Size: 0xf0000
15:20:54,4184774	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\System32\kernel32.dll	SUCCESS	Image Base: 0x140000, Image Size: 0xad000
15:20:54,4185955	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\System32\user32.dll	SUCCESS	Image Base: 0x140000, Image Size: 0x14e000
15:20:54,4188576	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\System32\wow64cpu.dll	SUCCESS	Image Base: 0x723f0000, Image Size: 0x8000
15:20:54,4201962	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\kernel32.dll	SUCCESS	Image Base: 0x76f70000, Image Size: 0xf0000
15:20:54,4204137	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\KernelBase.dll	SUCCESS	Image Base: 0x766e0000, Image Size: 0x176000
15:20:54,4223577	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\apphelp.dll	SUCCESS	Image Base: 0x73c10000, Image Size: 0x91000
15:20:54,4377490	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\shell32.dll	SUCCESS	Image Base: 0x74870000, Image Size: 0x13bf000
15:20:54,4380595	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\msvcrt.dll	SUCCESS	Image Base: 0x77060000, Image Size: 0xbe000
15:20:54,4384374	calcolatriceinnovativa.exe	4480	Thread Create		SUCCESS	Thread ID: 5604
15:20:54,4387231	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\windows.storage.dll	SUCCESS	Image Base: 0x75c80000, Image Size: 0x4dd000
15:20:54,4389856	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\combase.dll	SUCCESS	Image Base: 0x76280000, Image Size: 0x1ba000
15:20:54,4392172	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\rpcrt4.dll	SUCCESS	Image Base: 0x76880000, Image Size: 0xac000
15:20:54,4394587	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\sspicli.dll	SUCCESS	Image Base: 0x74230000, Image Size: 0x1e000
15:20:54,4396157	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\cryptbase.dll	SUCCESS	Image Base: 0x74220000, Image Size: 0xa000
15:20:54,4397909	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\bcryptprimitives.dll	SUCCESS	Image Base: 0x741c0000, Image Size: 0x59000
15:20:54,4399231	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\sechost.dll	SUCCESS	Image Base: 0x74820000, Image Size: 0x43000
15:20:54,4401745	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\advapi32.dll	SUCCESS	Image Base: 0x746d0000, Image Size: 0x7b000
15:20:54,4404389	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\shlwapi.dll	SUCCESS	Image Base: 0x75c30000, Image Size: 0x44000
15:20:54,4406545	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\gdi32.dll	SUCCESS	Image Base: 0x76a00000, Image Size: 0x14d000
15:20:54,4408698	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\user32.dll	SUCCESS	Image Base: 0x76590000, Image Size: 0x140000
15:20:54,4410690	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\kernel.appcore.dll	SUCCESS	Image Base: 0x76870000, Image Size: 0xc000
15:20:54,4413101	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\SHCore.dll	SUCCESS	Image Base: 0x74410000, Image Size: 0x8d000
15:20:54,4414964	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\powrprof.dll	SUCCESS	Image Base: 0x76440000, Image Size: 0x44000
15:20:54,4416987	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\profapi.dll	SUCCESS	Image Base: 0x76860000, Image Size: 0xf000
15:20:54,4473955	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\imm32.dll	SUCCESS	Image Base: 0x746a0000, Image Size: 0x2b000
15:20:54,4478492	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\msctf.dll	SUCCESS	Image Base: 0x76160000, Image Size: 0x120000
15:20:54,4505821	calcolatriceinnovativa.exe	4480	Thread Create		SUCCESS	Thread ID: 5156
15:20:54,4554134	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\ws2_32.dll	SUCCESS	Image Base: 0x76490000, Image Size: 0x5c000
15:20:54,4556164	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\nsi.dll	SUCCESS	Image Base: 0x766d0000, Image Size: 0x7000
15:20:54,4727760	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\mswsock.dll	SUCCESS	Image Base: 0x73970000, Image Size: 0x4e000
15:20:54,4741395	calcolatriceinnovativa.exe	4480	Load Image	C:\Windows\SysWOW64\rasadhlp.dll	SUCCESS	Image Base: 0x6def0000, Image Size: 0x8000
15:20:54,4775646	calcolatriceinnovativa.exe	4480	Thread Exit		SUCCESS	Thread ID: 3692, User Time: 0.0000000, Kernel Time: 0.0156250
15:20:54,4913372	calcolatriceinnovativa.exe	4480	Thread Exit		SUCCESS	Thread ID: 5156, User Time: 0.0000000, Kernel Time: 0.0000000
15:20:54,4914092	calcolatriceinnovativa.exe	4480	Thread Exit		SUCCESS	Thread ID: 5604, User Time: 0.0000000, Kernel Time: 0.0000000

Dalle attività di processi e thread si può osservare:

1. Il processo inizia caricando numerose librerie di sistema Windows (DLL).
2. Carica file da diverse cartelle di sistema, in particolare da **C:\Windows\System32** e **C:\Windows\SysWOW64**.
3. Durante l'esecuzione vengono creati diversi thread.
4. Tutte le operazioni mostrate risultano in **SUCCESS**, indicando che il malware è riuscito a eseguire le sue azioni senza errori.
5. Il processo termina dopo aver compiuto numerose operazioni di caricamento e creazione di thread.

Questo comportamento è tipico di un malware che cerca di:

1. Nascondersi come un'applicazione legittima (in questo caso una calcolatrice).
2. Caricare componenti di sistema per mascherare la propria attività o sfruttare vulnerabilità del sistema.
3. Creare thread multipli per eseguire diverse azioni contemporaneamente.
4. Potenzialmente stabilire persistenza nel sistema o eseguire altre attività malevole.

15:20:54.4154835	calcolatriceinnovativa.exe	4480	CreateFile	C:\Windows\Prefetch\CALCOLATRICEINNOVATIVA.EXE-51CE050C.pf	NAME NOT FOUND	Desired Access: Generic Read, Disposition: Open, Options: Synchronous IO Non-Alert, Attributes: n/a, ShareMode: None, AllocationSize: n/a
15:20:54.4157844	calcolatriceinnovativa.exe	4480	CreateFile	C:\Windows	SUCCESS	Desired Access: Execute/Traverse, Synchronize, Disposition: Open, Options: Directory, Synchronous IO Non-Alert, Attributes: n/a, ShareMode: R...
15:20:54.4177741	calcolatriceinnovativa.exe	4480	CreateFile	C:\Windows\System32\wow64log.dll	NAME NOT FOUND	Desired Access: Read Attributes, Disposition: Open, Options: Open Reparse Point, Attributes: n/a, ShareMode: Read, Write, Delete, AllocationSiz...
15:20:54.4186427	calcolatriceinnovativa.exe	4480	CreateFile	C:\Windows	SUCCESS	Desired Access: Read Attributes, Synchronize, Disposition: Open, Options: Synchronous IO Non-Alert, Attributes: n/a, ShareMode: Read, Write, D...
15:20:54.4186659	calcolatriceinnovativa.exe	4480	QueryNameInformationFile	C:\Windows	SUCCESS	Name: \Windows
15:20:54.4186720	calcolatriceinnovativa.exe	4480	CloseFile	C:\Windows	SUCCESS	
15:20:54.4200987	calcolatriceinnovativa.exe	4480	CreateFile	C:\Users\user\Desktop\Malware	SUCCESS	Desired Access: Execute/Traverse, Synchronize, Disposition: Open, Options: Directory, Synchronous IO Non-Alert, Attributes: n/a, ShareMode: R...
15:20:54.4212705	calcolatriceinnovativa.exe	4480	CreateFile	C:\Windows\SysWOW64\apphelp.dll	SUCCESS	Desired Access: Read Attributes, Disposition: Open, Options: Open Reparse Point, Attributes: n/a, ShareMode: Read, Write, Delete, AllocationSiz...
15:20:54.4213402	calcolatriceinnovativa.exe	4480	QueryBasicInformationFile	C:\Windows\SysWOW64\apphelp.dll	SUCCESS	CreationTime: 10/07/2015 13:00:23, LastAccessTime: 10/07/2015 13:00:23, LastWriteTime: 10/07/2015 13:00:23, ChangeTime: 09/07/2024 1...
15:20:54.4213463	calcolatriceinnovativa.exe	4480	CloseFile	C:\Windows\SysWOW64\apphelp.dll	SUCCESS	
15:20:54.4216450	calcolatriceinnovativa.exe	4480	CreateFile	C:\Windows\SysWOW64\apphelp.dll	SUCCESS	Desired Access: Read Data/List Directory, Execute/Traverse, Synchronize, Disposition: Open, Options: Synchronous IO Non-Alert, Non-Directory ...
15:20:54.4216694	calcolatriceinnovativa.exe	4480	CreateFileMapping	C:\Windows\SysWOW64\apphelp.dll	FILE LOCKED WITH ONLY READERS	SyncType: SyncTypeCreateSection, PageProtection:
15:20:54.4222168	calcolatriceinnovativa.exe	4480	CreateFileMapping	C:\Windows\SysWOW64\apphelp.dll	SUCCESS	SyncType: SyncTypeOther
15:20:54.4224758	calcolatriceinnovativa.exe	4480	CloseFile	C:\Windows\SysWOW64\apphelp.dll	SUCCESS	
15:20:54.4227421	calcolatriceinnovativa.exe	4480	CreateFile	C:\Users\user\Desktop\Malware\calcolatriceinnovativa.exe	SUCCESS	Desired Access: Read Control, Disposition: Open, Options: , Attributes: n/a, ShareMode: Read, Delete, AllocationSize: n/a, OpenResult: Opened
15:20:54.4227585	calcolatriceinnovativa.exe	4480	QuerySecurityFile	C:\Users\user\Desktop\Malware\calcolatriceinnovativa.exe	BUFFER OVERFLOW	Information: Owner
15:20:54.4227638	calcolatriceinnovativa.exe	4480	QuerySecurityFile	C:\Users\user\Desktop\Malware\calcolatriceinnovativa.exe	SUCCESS	Information: Owner
15:20:54.4227680	calcolatriceinnovativa.exe	4480	CloseFile	C:\Users\user\Desktop\Malware\calcolatriceinnovativa.exe	SUCCESS	
15:20:54.4228084	calcolatriceinnovativa.exe	4480	CreateFile	C:\Windows\SysWOW64\ntdll.dll	SUCCESS	Desired Access: Read Control, Disposition: Open, Options: , Attributes: n/a, ShareMode: Read, Delete, AllocationSize: n/a, OpenResult: Opened
15:20:54.4228431	calcolatriceinnovativa.exe	4480	QuerySecurityFile	C:\Windows\SysWOW64\ntdll.dll	BUFFER OVERFLOW	Information: Owner
15:20:54.4228469	calcolatriceinnovativa.exe	4480	QuerySecurityFile	C:\Windows\SysWOW64\ntdll.dll	SUCCESS	Information: Owner
15:20:54.4228511	calcolatriceinnovativa.exe	4480	CloseFile	C:\Windows\SysWOW64\ntdll.dll	SUCCESS	
15:20:54.4228911	calcolatriceinnovativa.exe	4480	CreateFile	C:\Windows\SysWOW64\kernel32.dll	SUCCESS	Desired Access: Read Control, Disposition: Open, Options: , Attributes: n/a, ShareMode: Read, Delete, AllocationSize: n/a, OpenResult: Opened
15:20:54.4229181	calcolatriceinnovativa.exe	4480	QuerySecurityFile	C:\Windows\SysWOW64\kernel32.dll	BUFFER OVERFLOW	Information: Owner
15:20:54.4229216	calcolatriceinnovativa.exe	4480	QuerySecurityFile	C:\Windows\SysWOW64\kernel32.dll	SUCCESS	Information: Owner
15:20:54.4229254	calcolatriceinnovativa.exe	4480	CloseFile	C:\Windows\SysWOW64\kernel32.dll	SUCCESS	
15:20:54.4229547	calcolatriceinnovativa.exe	4480	CreateFile	C:\Windows\SysWOW64\KernelBase.dll	SUCCESS	Desired Access: Read Control, Disposition: Open, Options: , Attributes: n/a, ShareMode: Read, Delete, AllocationSize: n/a, OpenResult: Opened
15:20:54.4229756	calcolatriceinnovativa.exe	4480	QuerySecurityFile	C:\Windows\SysWOW64\KernelBase.dll	BUFFER OVERFLOW	Information: Owner
15:20:54.4229791	calcolatriceinnovativa.exe	4480	QuerySecurityFile	C:\Windows\SysWOW64\KernelBase.dll	SUCCESS	Information: Owner
15:20:54.4229825	calcolatriceinnovativa.exe	4480	CloseFile	C:\Windows\SysWOW64\KernelBase.dll	SUCCESS	
15:20:54.4230195	calcolatriceinnovativa.exe	4480	CreateFile	C:\Windows\AppPatch\sysmain.sdb	SUCCESS	Desired Access: Generic Read, Disposition: Open, Options: Synchronous IO Non-Alert, Non-Directory File, Attributes: N, ShareMode: Read, Allocat...
15:20:54.4230366	calcolatriceinnovativa.exe	4480	QueryStandardInformationFile	C:\Windows\AppPatch\sysmain.sdb	SUCCESS	AllocationSize: 3.735.552, EndOfFile: 3.734.416, NumberOfLinks: 2, DeletePending: False, Directory: False
15:20:54.4230416	calcolatriceinnovativa.exe	4480	QueryStandardInformationFile	C:\Windows\AppPatch\sysmain.sdb	SUCCESS	AllocationSize: 3.735.552, EndOfFile: 3.734.416, NumberOfLinks: 2, DeletePending: False, Directory: False
15:20:54.4230465	calcolatriceinnovativa.exe	4480	CreateFileMapping	C:\Windows\AppPatch\sysmain.sdb	FILE LOCKED WITH ONLY READERS	SyncType: SyncTypeCreateSection, PageProtection:
15:20:54.4230507	calcolatriceinnovativa.exe	4480	QueryStandardInformationFile	C:\Windows\AppPatch\sysmain.sdb	SUCCESS	AllocationSize: 3.735.552, EndOfFile: 3.734.416, NumberOfLinks: 2, DeletePending: False, Directory: False
15:20:54.4230579	calcolatriceinnovativa.exe	4480	CreateFileMapping	C:\Windows\AppPatch\sysmain.sdb	SUCCESS	SyncType: SyncTypeOther
15:20:54.4232640	calcolatriceinnovativa.exe	4480	CreateFile	C:\Users\user\Desktop\Malware\calcolatriceinnovativa.exe	SUCCESS	Desired Access: Generic Read, Disposition: Open, Options: Synchronous IO Non-Alert, Non-Directory File, Attributes: N, ShareMode: Read, Allocat...
15:20:54.4233136	calcolatriceinnovativa.exe	4480	QuerySecurityFile	C:\Users\user\Desktop\Malware\calcolatriceinnovativa.exe	SUCCESS	Information: Owner, Group, DACL, SACL, Label
15:20:54.4233482	calcolatriceinnovativa.exe	4480	CreateFile	C:\Windows\AppPatch\sysmain.sdb	SUCCESS	Desired Access: Generic Read, Disposition: Open, Options: Synchronous IO Non-Alert, Non-Directory File, Attributes: N, ShareMode: Read, Allocat...
15:20:54.4233627	calcolatriceinnovativa.exe	4480	QueryBasicInformationFile	C:\Windows\AppPatch\sysmain.sdb	SUCCESS	CreationTime: 10/07/2015 13:00:34, LastAccessTime: 10/07/2015 13:00:34, LastWriteTime: 10/07/2015 13:00:34, ChangeTime: 09/07/2024 1...
15:20:54.4233673	calcolatriceinnovativa.exe	4480	CloseFile	C:\Windows\AppPatch\sysmain.sdb	SUCCESS	
15:20:54.4234004	calcolatriceinnovativa.exe	4480	CreateFile	C:\Windows\AppPatch\apppatch64\aysmain.sdb	SUCCESS	Desired Access: Generic Read, Disposition: Open, Options: Synchronous IO Non-Alert, Non-Directory File, Attributes: N, ShareMode: Read, Allocat...
15:20:54.4234126	calcolatriceinnovativa.exe	4480	QueryBasicInformationFile	C:\Windows\AppPatch\apppatch64\aysmain.sdb	SUCCESS	CreationTime: 10/07/2015 13:00:21, LastAccessTime: 10/07/2015 13:00:21, LastWriteTime: 10/07/2015 13:00:21, ChangeTime: 09/07/2024 1...
15:20:54.4234156	calcolatriceinnovativa.exe	4480	CloseFile	C:\Windows\AppPatch\apppatch64\aysmain.sdb	SUCCESS	
15:20:54.4234267	calcolatriceinnovativa.exe	4480	QueryBasicInformationFile	C:\Users\user\Desktop\Malware\calcolatriceinnovativa.exe	SUCCESS	CreationTime: 22/07/2024 12:08:38, LastAccessTime: 22/07/2024 12:08:38, LastWriteTime: 22/07/2024 12:00:44, ChangeTime: 22/07/2024 1...

Dalle attività sul file system si può osservare:

1. **Esecuzione e caricamento del malware**
   - Il malware viene eseguito dalla cartella C:\Users\user\Desktop\Malware.
   - Tenta di accedere a vari file di sistema e DLL, probabilmente per caricarli in memoria.
2. **Accesso ai file di sistema**
   - Accede a numerose DLL di sistema nella cartella C:\Windows\SysWOW64\, tra cui apphelp.dll, kernel32.dll, user32.dll, ws2_32.dll (utilizzata per funzionalità di rete) e altre.
   - Questo comportamento potrebbe servire a nascondere la propria attività oppure a utilizzare funzioni di sistema legittime.
3. **Operazioni di lettura e mapping**
   - Esegue operazioni di lettura su alcune DLL, come apphelp.dll e ws2_32.dll.
   - Crea file mapping per diverse DLL, operazione che potrebbe indicare tentativi di iniezione di codice o manipolazione della memoria.
4. **Interrogazioni di sicurezza**
   - Esegue numerose query di sicurezza sui file.
   - Questo potrebbe indicare tentativi di identificare le impostazioni di sicurezza del sistema o di individuare eventuali vulnerabilità.
5. **Attività di rete**
   - Il caricamento di ws2_32.dll e mswsock.dll suggerisce che il malware potrebbe tentare di stabilire connessioni di rete.
6. **Persistenza e nascondimento**
   - Nei log forniti non sono presenti chiari segni di tentativi di persistenza nel sistema.
   - Tuttavia, potrebbe trattarsi di una fase successiva non catturata nel log analizzato.
7. **Possibile keylogging o cattura dell'input**
   - Il caricamento di imm32.dll e user32.dll potrebbe indicare tentativi di intercettare l'input dell'utente.
8. **Esplorazione del sistema**
   - Le numerose query sui file di sistema suggeriscono che il malware stia raccogliendo informazioni sul sistema host.
9. **Possibile offuscamento**
   - L'accesso a molte DLL di sistema potrebbe essere un tentativo di mascherare la propria natura, mescolando la sua attività con operazioni di sistema normali.

Conclusione

Questo malware sembra trovarsi nella fase iniziale di esecuzione, durante la quale carica diverse librerie di sistema e si prepara potenzialmente a svolgere attività malevole, come stabilire connessioni di rete, intercettare l'input dell'utente o manipolare il sistema.

Il suo comportamento suggerisce che potrebbe trattarsi di un **trojan** o di un **payload iniziale**, progettato per stabilire una presenza nel sistema prima di scaricare o attivare funzionalità più dannose.